PILOT: an offline routing study

Team 8 | HCMUS course project | Complete declared local experiments

8 model arms / 3 seeds / 1,098 budget runs / No paid generation calls

The question

How much recorded reward can a router deliver under a limited estimated budget, and how does preference initialization change its behavior?

Offline workflow

Human preferences > local BGE-M3 embeddings > seeded projection and arm priors > chosen-arm learning feedback > frozen budget evaluation.

Data and controls

RouterBench: 32,270 learning and 3,227 deployment queries. MMLU subset: 12,369 and 1,275. Tuning is separate. All methods share each domain's pool, order, budgets and estimated-cost rules.

RouterBench result

At 0.5 of the reference budget, PILOT delivered 0.676 reward per requested query with 100.0% coverage. The highest mean at this budget was LinUCB (0.706). These are descriptive results on fixed streams.

MMLU result

At 0.5 of the reference budget, PILOT delivered 0.667 reward per requested query with 100.0% coverage. The highest mean at this budget was ExploreOnly (0.690). These are descriptive results on fixed streams.

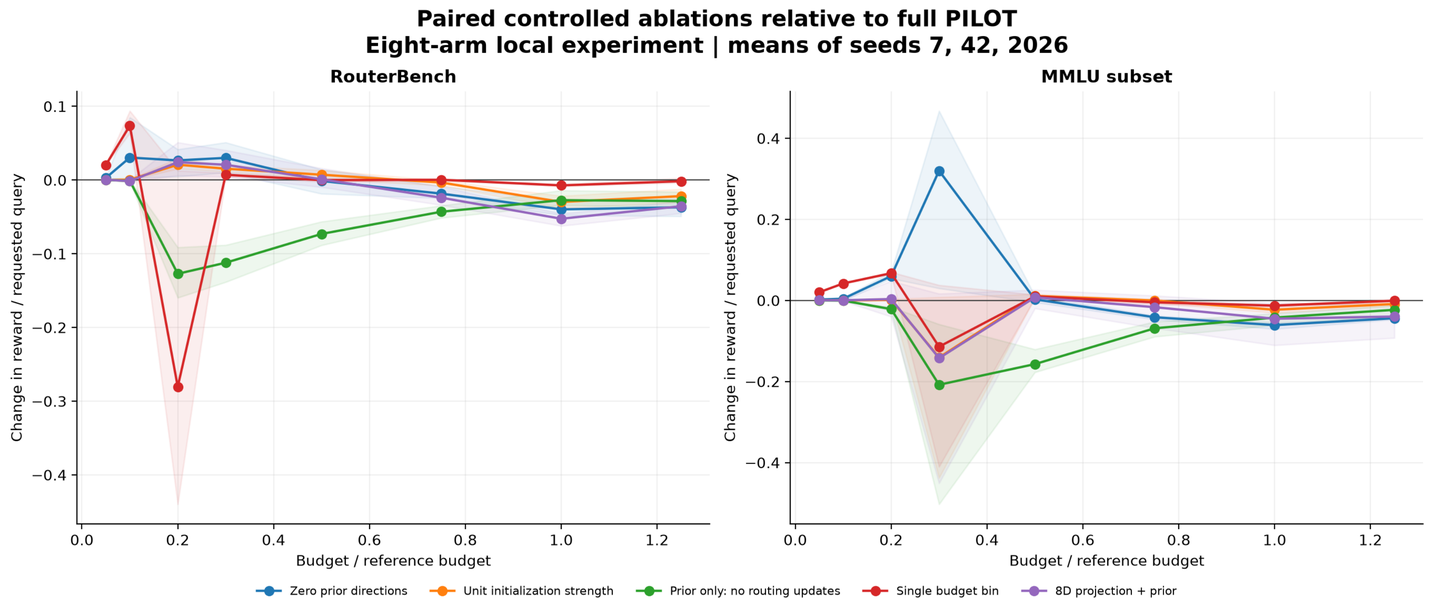
Evidence and limits

All 108 method/variant groups independently replayed. 12 conditions exceeded the budget in recorded cost despite estimated affordability. Three seeds provide descriptive ranges only. BGE-M3, model pool and source details differ from the paper; full eleven-arm equivalence remains unresolved.

Budgeted service utility - every primary method



Five paired ablations - including unfavorable results



Reference: Panda et al., Findings of EMNLP 2025 | DOI 10.18653/v1/2025.findings-emnlp.1301 | Local experiment, not paper-equivalent